

Steven Dang

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EDUCATION

University of Texas at Arlington — 3.80/4.0 GPA

Bachelor of Science in Computer Science

Arlington, TX

May 2027

EXPERIENCE

Undergraduate Research Lead

Sept 2025 – December 2025

University of Texas at Arlington

Arlington, TX

- Led simulation development for a Hodgkin–Huxley literature review, implementing ion-channel and membrane-dynamics models based on primary research papers.
- Introduced Brian2 as the primary simulation framework, replacing a Matplotlib-only workflow that lacked support for large parameter sweeps and multi-variable differential equation modeling.
- Acted as a technical bridge between theory and implementation: distilled complex neurophysiological concepts into actionable modeling tasks for the research team.

Owner & Operator — Popup Café Business

May 2025 – Present

Koicha LLC

Fort Worth, TX

- Founded and operated a pop-up café business, managing end-to-end operations including menu development, staffing, inventory planning, and on-site service.
- Built a custom Google Apps Script–based order-management system that processed 5,000+ customer orders, automating logging, receipt generation, and daily reporting.
- Designed internal tools integrating Google Sheets and Apps Script to streamline order tracking, inventory updates, and real-time metrics dashboards.

National Community College Aerospace Scholar

May 2024 – Dec 2024

NASA NCAS

Dallas, TX

- Produced LaTeX-formatted technical documents and infographics summarizing NASA’s engineering standards and mission directives.
- Practiced translating complex aerospace documentation into clear, human-readable materials for student engineering teams.
- Built early experience with structured technical writing, documentation workflows, and visual communication.

PROJECTS

OpenLedger POS - Open Source Project | *Python, Flask, SQLAlchemy, PostgreSQL, React, TailwindCSS*

- Developed a self-hostable point-of-sale management system to record and manage drink orders, designed as an open alternative to centralized SaaS POS platforms
- Designed RESTful endpoints (GET/POST/PUT/DELETE) for user authentication, order creation, and inventory management, exchanging JSON over HTTP between React frontend and Flask backend.
- Designed relational schemas in PostgreSQL and mapped them using SQLAlchemy ORM to enforce structured relationships and data integrity
- Built a React frontend communicating with the backend via JSON HTTP requests for real-time order entry and management

Particle Simulator | *Python, NumPy, Matplotlib*

- Designed a physics-based particle simulation engine modeling forces, velocity, and acceleration via numerical integration methods (Modified Euler).
- Optimized computation for large particle sets and built dynamic Matplotlib visualizations to analyze motion patterns in real time.
- Validated simulation results against theoretical physics models to ensure numerical stability and correctness.

TECHNICAL SKILLS

Languages: Java, Python, C, C#, SQL (Postgres, SQLAlchemy, SQLite), JavaScript, HTML/CSS

Frameworks: React, Node.js, Next.js, Flask, WordPress

Developer Tools: Git, Vite, Docker, PostgreSQL, Google Cloud Platform, Jupyter Notebooks, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Google App Script

Libraries: pandas, NumPy, Matplotlib, XGBoost, Scikit-learn, seaborn, Brian2, Grommet